



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

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Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	Vedant Kishor Mhatre
Roll No. :	72
Experiment No.:	4
Title of the Experiment:	Perform Statistical Inference, Estimation, and Hypothesis Testing on Real-World Datasets
Date of Performance:	
Date of Submission:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty :
Signature :
Date :



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Aim: To apply statistical estimation and hypothesis testing techniques to the Pima Indians Diabetes Dataset and draw conclusions about population characteristics using sample data.

Objective

After completing this experiment, the student will be able to:

1. Apply estimation and hypothesis testing techniques to analyze population characteristics using sample data.
2. Interpret statistical test results and make data-driven conclusions based on the obtained evidence.

Tools Required

1. Python 3.x
2. Google Colab / Jupyter Notebook
3. Pandas
4. NumPy
5. SciPy
6. Matplotlib
7. Seaborn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for 768 female patients

Procedure

1. Load the Pima Indians Diabetes Dataset.
2. Select appropriate variables and divide the data into suitable groups where required.
3. Calculate point estimates such as sample mean and sample proportion.
4. Construct confidence intervals for selected population parameters.
5. Formulate null and alternative hypotheses for selected research questions.
6. Select and apply an appropriate statistical test, such as:



- a. One-sample t-test
 - b. Two-sample t-test
 - c. Chi-square test
7. Calculate the test statistic and p-value.
 8. Compare the p-value with the selected significance level.
 9. Make a statistical decision and interpret the result in the context of the dataset.

Description of the Experiment

Statistical inference allows conclusions about a population to be drawn using information from a sample.

The general process is: Sample Data, Estimate Population Parameter, Construct Confidence Interval, Formulate Hypotheses, Perform Statistical Test, and Interpret Results

For example, a sample of patients can be used to investigate whether the average glucose level differs from a specified reference value or whether diabetes outcome is associated with another categorical variable.

Detailed Description of Statistical Methods

Point Estimation

A sample statistic is used to estimate an unknown population parameter.

For example:

$$\bar{x}$$

is used to estimate the population mean μ .

A sample proportion is:

$$\hat{p} = \frac{x}{n}$$

where x is the number of observations with a particular characteristic.



Confidence Interval

A confidence interval provides a range of plausible values for a population parameter.

For a mean:

$$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

The confidence interval should be interpreted in the context of the population parameter being estimated.

Hypothesis Testing

A hypothesis test evaluates a claim about a population.

Null Hypothesis

$$H_0$$

represents the initial assumption.

Alternative Hypothesis

$$H_1$$

represents the claim being investigated.

A significance level such as:

$$\alpha = 0.05$$

may be used.

Decision Rule

- If $p - value < \alpha$, reject H_0 .
- If $p - value \geq \alpha$, fail to reject H_0 .



t-Test

A t-test can be used to compare means.

Examples:

- Test whether the mean glucose level differs from a specified value.
- Compare the mean BMI of two groups.

Chi-Square Test

The chi-square test can be used to test the association between categorical variables.

For example:

Diabetes Outcome and Gender

where applicable.

Code:

```
import pandas as pd

import numpy as np

import scipy.stats as stats

import matplotlib.pyplot as plt

import seaborn as sns

url =
"https://raw.githubusercontent.com/jbrownlee/Datasets/master/pima-indians-diabetes.data.csv"
"columns = [

    "Pregnancies", "Glucose", "BloodPressure", "SkinThickness",

    "Insulin", "BMI", "DiabetesPedigreeFunction", "Age", "Outcome"

]
```



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```
df = pd.read_csv(url, names=columns)

print("First 5 rows:")

print(df.head())

print("\nDataset Shape:", df.shape)

print("\nDataset Information:")

print(df.info())

print("\nDescriptive Statistics:")

print(df.describe())

sample_mean_glucose = df["Glucose"].mean()

print("\nSample Mean Glucose:", sample_mean_glucose)

diabetic_count = df["Outcome"].sum()

total_count = len(df)

sample_proportion = diabetic_count / total_count

print("Number of diabetic patients:", diabetic_count)

print("Total patients:", total_count)

print("Sample Proportion:", sample_proportion)

print("Sample Proportion (%):", sample_proportion * 100)

n = len(df)

mean = df["Glucose"].mean()

std = df["Glucose"].std()
```



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```
confidence = 0.95
```

```
alpha = 1 - confidence
```

```
t_critical = stats.t.ppf(1 - alpha/2, df=n-1)
```

```
margin_error = t_critical * (std / np.sqrt(n))
```

```
lower_ci = mean - margin_error
```

```
upper_ci = mean + margin_error
```

```
print("\n95% Confidence Interval for Mean Glucose:")
```

```
print("Lower Limit:", lower_ci)
```

```
print("Upper Limit:", upper_ci)
```

```
p = sample_proportion
```

```
se_proportion = np.sqrt((p * (1 - p)) / n)
```

```
z_critical = stats.norm.ppf(1 - alpha/2)
```

```
lower_prop_ci = p - z_critical * se_proportion
```

```
upper_prop_ci = p + z_critical * se_proportion
```

```
print("\n95% Confidence Interval for Diabetes Proportion:")
```

```
print("Lower Limit:", lower_prop_ci)
```

```
print("Upper Limit:", upper_prop_ci)
```

```
reference_value = 120
```

```
t_statistic, p_value = stats.ttest_1samp(
```

```
    df["Glucose"],
```

```
    reference_value
```



```
)

print("\nOne-Sample t-Test")

print("H0: Mean glucose = 120")

print("H1: Mean glucose != 120")

print("t-statistic:", t_statistic)

print("p-value:", p_value)

alpha = 0.05

if p_value < alpha:

    print("Decision: Reject H0")

    print("Conclusion: There is significant evidence that the mean glucose level differs from
120 mg/dL.")

else:

    print("Decision: Fail to reject H0")

    print("Conclusion: There is insufficient evidence to conclude that the mean glucose level
differs from 120 mg/dL.")

bmi_non_diabetic = df[df["Outcome"] == 0]["BMI"]

bmi_diabetic = df[df["Outcome"] == 1]["BMI"]

t_stat_bmi, p_value_bmi = stats.ttest_ind(

    bmi_diabetic,

    bmi_non_diabetic,

    equal_var=False

)
```




```
print("\nTwo-Sample t-Test for BMI")

print("H0: Mean BMI of diabetic and non-diabetic patients is equal")

print("H1: Mean BMI of diabetic and non-diabetic patients is different")

print("Mean BMI - Non-diabetic:", bmi_non_diabetic.mean())

print("Mean BMI - Diabetic:", bmi_diabetic.mean())

print("t-statistic:", t_stat_bmi)

print("p-value:", p_value_bmi)

if p_value_bmi < alpha:

    print("Decision: Reject H0")

    print("Conclusion: There is a significant difference in BMI between diabetic and
non-diabetic patients.")

else:

    print("Decision: Fail to reject H0")

    print("Conclusion: There is insufficient evidence of a difference in BMI.")

df["BMI_Category"] = pd.cut(

    df["BMI"],

    bins=[0, 25, 30, np.inf],

    labels=["Normal", "Overweight", "Obese"]

)

contingency_table = pd.crosstab(

    df["BMI_Category"],
```



```
df["Outcome"]

)

print("\nContingency Table:")

print(contingency_table)

chi2, p_chi, dof, expected = stats.chi2_contingency(

    contingency_table

)

print("\nChi-Square Test")

print("H0: BMI category and diabetes outcome are independent")

print("H1: BMI category and diabetes outcome are associated")

print("Chi-square statistic:", chi2)

print("Degrees of freedom:", dof)

print("p-value:", p_chi)

if p_chi < alpha:

    print("Decision: Reject H0")

    print("Conclusion: BMI category and diabetes outcome are significantly associated.")

else:

    print("Decision: Fail to reject H0")

    print("Conclusion: There is insufficient evidence of an association.")

sns.histplot(df["Glucose"], kde=True)

plt.axvline(120, linestyle="--", label="Reference = 120")
```



```
plt.title("Distribution of Glucose Levels")

plt.xlabel("Glucose")

plt.ylabel("Frequency")

plt.legend()

plt.show()

plt.figure(figsize=(8, 5))

sns.boxplot(x="Outcome", y="BMI", data=df)

plt.title("BMI Distribution by Diabetes Outcome")

plt.xlabel("Diabetes Outcome (0 = No, 1 = Yes)")

plt.ylabel("BMI")

plt.show()

plt.figure(figsize=(8, 5))

sns.countplot(x="BMI_Category", hue="Outcome", data=df)

plt.title("BMI Category vs Diabetes Outcome")

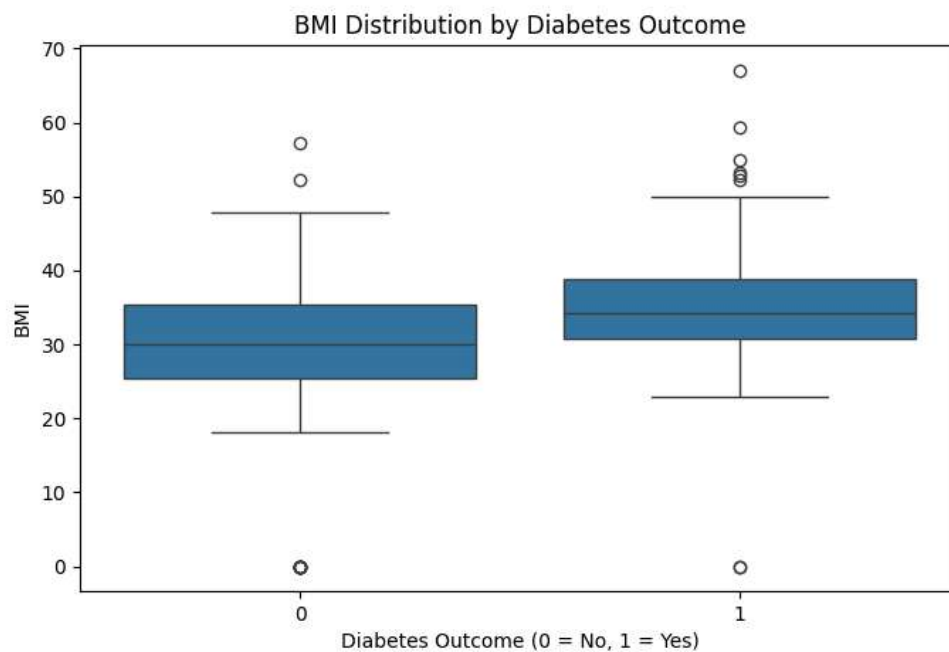
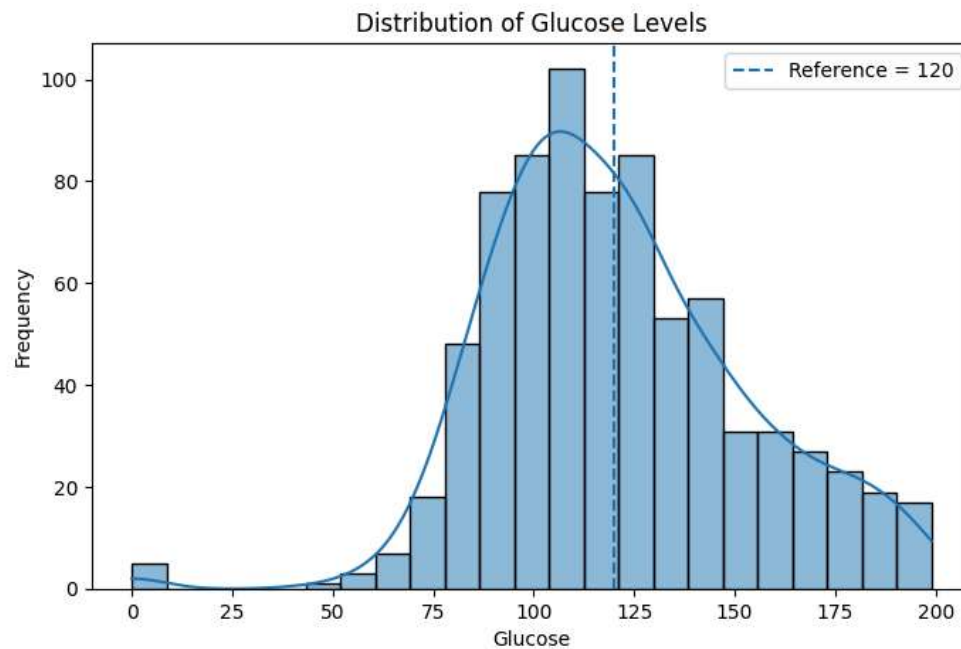
plt.xlabel("BMI Category")

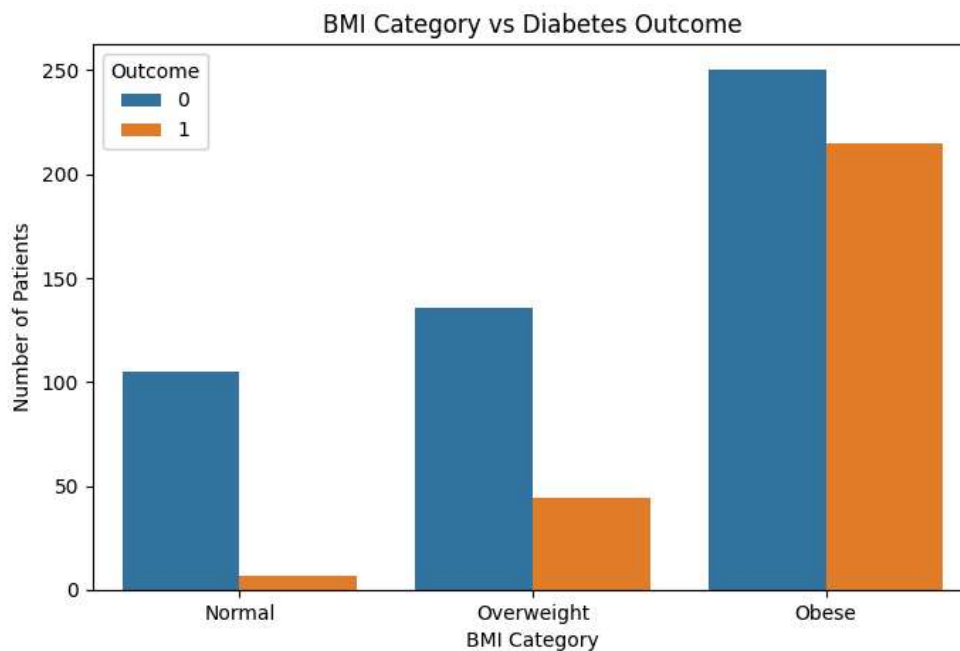
plt.ylabel("Number of Patients")

plt.show()
```



Output:





First 5 rows:

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1

Dataset Shape: (768, 9)

Dataset Information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 768 entries, 0 to 767

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	Pregnancies	768 non-null	int64
1	Glucose	768 non-null	int64
2	BloodPressure	768 non-null	int64
3	SkinThickness	768 non-null	int64
4	Insulin	768 non-null	int64
5	BMI	768 non-null	float64
6	DiabetesPedigreeFunction	768 non-null	float64
7	Age	768 non-null	int64
8	Outcome	768 non-null	int64

dtypes: float64(2), int64(7)

memory usage: 54.1 KB

None



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```
Descriptive Statistics:
Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin  BMI  DiabetesPedigreeFunction  Age  Outcome
count  768.000000  768.000000  768.000000  768.000000  768.000000  768.000000  768.000000  768.000000  768.000000
mean    3.845052  120.894531  69.105469  20.536458  79.799479  31.992578  0.471876  33.240885  0.348958
std     3.369578  31.972618  19.355807  15.952218  115.244002  7.884160  0.331329  11.760232  0.476951
min     0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.078000  21.000000  0.000000
25%     1.000000  99.000000  62.000000  0.000000  0.000000  27.300000  0.243750  24.000000  0.000000
50%     3.000000  117.000000  72.000000  23.000000  30.500000  32.000000  0.372500  29.000000  0.000000
75%     6.000000  140.250000  80.000000  32.000000  127.250000  36.600000  0.626250  41.000000  1.000000
max     17.000000  199.000000  122.000000  99.000000  846.000000  67.100000  2.420000  81.000000  1.000000

Sample Mean Glucose: 120.89453125
Number of diabetic patients: 268
Total patients: 768
Sample Proportion: 0.3489583333333333
Sample Proportion (%): 34.89583333333333

95% Confidence Interval for Mean Glucose:
Lower Limit: 118.62972245094552
Upper Limit: 123.15934004905448

95% Confidence Interval for Diabetes Proportion:
Lower Limit: 0.3152483405027714
Upper Limit: 0.38266832616389523

One-Sample t-Test
H0: Mean glucose = 120
H1: Mean glucose != 120
t-statistic: 0.7753502397592414
p-value: 0.4383717099127056
Decision: Fail to reject H0
Conclusion: There is insufficient evidence to conclude that the mean glucose level differs from 120 mg/dL.
```

```
Two-Sample t-Test for BMI
H0: Mean BMI of diabetic and non-diabetic patients is equal
H1: Mean BMI of diabetic and non-diabetic patients is different
Mean BMI - Non-diabetic: 30.3042
Mean BMI - Diabetic: 35.14253731343284
t-statistic: 8.619316881357946
p-value: 6.566237624708332e-17
Decision: Reject H0
Conclusion: There is a significant difference in BMI between diabetic and non-diabetic patients.

Contingency Table:
Outcome      0      1
BMI_Category
Normal        105     7
Overweight    136    44
Obese         250   215

Chi-Square Test
H0: BMI category and diabetes outcome are independent
H1: BMI category and diabetes outcome are associated
Chi-square statistic: 75.17171758432994
Degrees of freedom: 2
p-value: 4.749730068434557e-17
Decision: Reject H0
Conclusion: BMI category and diabetes outcome are significantly associated.
PS C:\Users\student\vedant\exp4>
```



Conclusion

Statistical inference, estimation, and hypothesis testing techniques were applied to the Pima Indians Diabetes Dataset. Sample statistics were used to estimate population characteristics, confidence intervals were constructed, and hypothesis tests were performed to determine whether sufficient statistical evidence existed to support specific claims.

Questions to be Answered by Students

1. What population parameter was estimated in the experiment, and what was its estimated value?

The main population parameter estimated was the **population mean glucose level**.

The sample mean is: $\bar{x}=120.89$ mg/dL

Therefore, the estimated population mean glucose level is approximately **120.89 mg/dL**.

We also estimated the population proportion of diabetic patients: $p^*=768268=0.349$

Therefore, approximately **34.9%** of the patients in the dataset have a positive diabetes outcome.

2. What confidence interval was obtained, and how should it be interpreted?

A **95% confidence interval** was calculated for the population mean glucose level.

Approximately: $CI_{95\%}=(118.63, 123.16)$

So, the 95% confidence interval for the population mean glucose level is approximately: **118.63 mg/dL to 123.16 mg/dL**.

Interpretation: We are 95% confident that the true population mean glucose level represented by this sample lies between approximately **118.63 and 123.16 mg/dL**.

For the diabetes proportion, the approximate 95% confidence interval is: (0.315, 0.383) or approximately **31.5% to 38.3%**.



3. What hypotheses were formulated for the selected statistical test?

For the **one-sample t-test**, we tested whether the average glucose level differs from 120 mg/dL.

Null hypothesis: $H_0: \mu = 120$

The population mean glucose level is 120 mg/dL.

Alternative hypothesis: $H_1: \mu \neq 120$

The population mean glucose level is different from 120 mg/dL.

The significance level was: $\alpha = 0.05$

The test gives approximately: $t = 0.775$ and $p \approx 0.438$

Since: $p > 0.05$ we **fail to reject H_0** .

4. Based on the p-value, what statistical decision was made?

For the one-sample t-test:

Test	p-value	α	Decision
One-sample t-test	≈ 0.438	0.05	Fail to reject H_0

Since **0.438 > 0.05**, the null hypothesis is **not rejected**.

Therefore, the experiment does not provide sufficient evidence that the population mean glucose level is different from 120 mg/dL.